Solution to Exercise 08 — Analysis on Manifolds, 80416

June 9, 2026

Question 1

Subquestion 1

We will use Guldin's theorem to calculate the area of $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$.

Solution: We follow the recitation's definition, let $\gamma : (a, b) \to \mathbb{R}^2$ be a smooth map denoted by $\gamma(t) = (x(t), z(t))$.
We need to ensure that $x(t) > 0$ for all $t \in (a, b)$ so $x(t) = \cos(t), z(t) = \sin(t)$ for $t \in (-\frac{\pi}{2}, \frac{\pi}{2})$.
Our surface of revolution is the sphere and it's obtained by γ as

$$M = \left\{(x(t)\cos(\theta), x(t)\sin(\theta), z(t)) \mid t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \theta \in [0, 2\pi]\right\}$$

we have

$$\|\dot{\gamma}(t)\| = \sqrt{(-\sin(t))^2 + (\cos(t))^2} = \sqrt{\sin^2(t) + \cos^2(t)} = \sqrt{1} = 1$$

Hence

$$l(\gamma) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \|\dot{\gamma}(t)\| \, \mathrm{d}t = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 \, \mathrm{d}t = \pi$$

Now we need to find the center of mass a_1.
Let a be the center of mass of $\gamma(t)$ in $\mathbb{R}^3$ so

$$a_1 = \frac{1}{l(\gamma)} \int_\gamma x \, \mathrm{d}\sigma$$

We know that

$$\int_\gamma x \, \mathrm{d}\sigma = \int_a^b x(t) \|\dot{\gamma}(t)\| \, \mathrm{d}t$$

Hence

$$a_1 = \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(t) \, \mathrm{d}t = \frac{1}{\pi} \cdot [\sin(t)]_{t=-\frac{\pi}{2}}^{t=\frac{\pi}{2}} = \frac{1}{\pi}(1 - (-1)) = \frac{2}{\pi}$$

Now we apply Guldin's theorem

$$\mathrm{Area}(M) = l(\gamma) \cdot 2\pi |a_1|$$

So in our case

$$\mathrm{Area}(M) = \pi \cdot 2\pi \cdot \frac{2}{\pi} = 4\pi$$

□

Subquestion 2

We will use Guldin's theorem to calculate the area of $C = \{(x, y, z) \in \mathbb{R}^3 \mid z^2 = a^2(x^2 + y^2), 0 < z < h \ a, h > 0\}$.

Solution: To find the generating curve γ in the xz-plane we set $y = 0$ so from the equation of the cone we have $z^2 = a^2 x^2$. Since $a > 0$ and $x(t) > 0$ we can take root and get that $z = ax$.
This is a line segment that can be parametrize as $x(t) = t, z(t) = at$ and we have constraint on z with h so

$$0 < z < h \Longleftrightarrow 0 < at < h \Longrightarrow 0 < t < \frac{h}{a}$$

So we define $\gamma : \left(0, \frac{h}{a}\right) \to \mathbb{R}^2$ be the smooth map $\gamma(t) = (x(t), z(t))$ with $x(t) = t, z(t) = at$ and $t \in \left(0, \frac{h}{a}\right)$.
So we can define

$$M = \left\{(t\cos(\theta), t\sin(\theta), at) \mid t \in \left(0, \frac{h}{a}\right), \theta \in [0, 2\pi]\right\}$$

Same as before

$$\dot{\gamma}(t) = (x'(t), z'(t)) = (1, a)$$
$$\|\dot{\gamma}(t)\| = \sqrt{1^2 + a^2} = \sqrt{1 + a^2}$$
$$l(\gamma) = \int_0^{\frac{h}{a}} \|\dot{\gamma}(t)\| = \int_0^{\frac{h}{a}} \sqrt{1 + a^2}\,\mathrm{d}t = \sqrt{1 + a^2}[t]_{t=0}^{t=\frac{h}{a}} = \frac{h}{a}\sqrt{1 + a^2}$$
$$a_1 = \frac{1}{l(\gamma)} \int_\gamma x\,\mathrm{d}\sigma = \frac{1}{l(\gamma)} \int_0^{\frac{h}{a}} t\|\dot{\gamma}(t)\|\,\mathrm{d}t = \frac{1}{\frac{h}{a}\sqrt{1 + a^2}} \int_0^{\frac{h}{a}} t\sqrt{1 + a^2}\,\mathrm{d}t = \frac{a}{2h}[t^2]_{t=0}^{t=\frac{h}{a}} = \frac{a}{h} \cdot \frac{h^2}{2a^2} = \frac{h}{2a}$$

Since $a, h > 0$ we have that $a_1 > 0$ so $|a_1| = a_1$ and from Guldin's theorem we get that

$$\mathrm{Area}(M) = \left(\frac{h}{a}\sqrt{1 + a^2}\right) \cdot 2\pi \cdot \frac{h}{2a} = \frac{2\pi h^2\sqrt{1 + a^2}}{2a^2} = \frac{\pi h^2\sqrt{1 + a^2}}{a^2}$$

□

Question 2

Consider

$$M = \left\{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 1, \sqrt{x^2+y^2} \le z^2\right\}$$
$$F(x,y,z) = \left(\frac{x^2}{2} + xy - 4xz + \cos(z), -\frac{y^2}{2} - xy + x^2 + z^3, 2z^2 + x\right)$$

We will calculate the flux of F through M with respect to the unit normal pointing upward.

Solution: Let $F = (P, Q, R)$ and we compute the divergence of F

$$P_x = \frac{\partial}{\partial x}\left(\frac{x^2}{2} + xy - 4xz + \cos(z)\right) = x + y - z$$
$$Q_y = \frac{\partial}{\partial y}\left(-\frac{y^2}{2} - xy + x^2 + z^3\right) = -y - x$$
$$R_z = \frac{\partial}{\partial z}(2z^2 + x) = 4z$$

By definition

$$\operatorname{div} F = P_x + Q_y + R_z = (x + y - z) + (-y - x) + 4z = 0$$

We need to find the boundary of M.
Let $r = \sqrt{x^2+y^2}$ and our conditions becomes $r^2 + z^2 = 1$ and $r \le z^2$ so $r \le 1 - r^2$ means that $r^2 + r - 1 \le 0$ and since $r \ge 0$ possible range is $0 \le r \le \frac{\sqrt{5}-1}{2}$ so let $R = \frac{\sqrt{5}-1}{2}$.
On the boundary we have $z^2 = 1 - R^2 = R$ so $z = \sqrt{R}$ and $z = -\sqrt{R}$ thus M consists of two disjoint parts M_1 where $z > 0$ and M_2 where $z < 0$.
Same as the recitation, we denote

$$D_1 = \left\{(x,y,z) \in \mathbb{R}^3 \mid z = \sqrt{R}, x^2+y^2 < R^2\right\}$$
$$V_1 = \left\{(x,y,z) \in \mathbb{R}^3 \mid \sqrt{R} < z < \sqrt{1-x^2-y^2}, x^2+y^2 < R^2\right\}$$
$$D_2 = \left\{(x,y,z) \in \mathbb{R}^3 \mid z = -\sqrt{R}, x^2+y^2 < R^2\right\}$$
$$V_2 = \left\{(x,y,z) \in \mathbb{R}^3 \mid -\sqrt{1-x^2-y^2} < z \leftarrow \sqrt{R}, x^2+y^2 < R^2\right\}$$

By the divergence theorem

$$\int_{M_1 \cup D_1} \langle F, N\rangle \,\mathrm{d}\sigma = \int_{M_1} \langle F, N\rangle \,\mathrm{d}\sigma = \int_{D_1} \langle F, N\rangle \,\mathrm{d}\sigma = \int_{V_1} \operatorname{div} F \,\mathrm{d}x \,\mathrm{d}y \,\mathrm{d}z = 0$$
$$\int_{M_2 \cup D_2} \langle F, N\rangle \,\mathrm{d}\sigma = \int_{M_2} \langle F, N\rangle \,\mathrm{d}\sigma = \int_{D_2} \langle F, N\rangle \,\mathrm{d}\sigma = \int_{V_2} \operatorname{div} F \,\mathrm{d}x \,\mathrm{d}y \,\mathrm{d}z = 0$$

Let $N = (0, 0, -1)$. Let $p \in D_1$ and let $\gamma : (-\varepsilon, \varepsilon) \to D_1$ be a differentiable path such that $\gamma(0) = p$.
Since $Im\,\gamma \subseteq D_1$ we can write it as $\gamma(t) = \left(x(t), y(t), \sqrt{R}\right)$ thus

$$\langle \dot{\gamma}(0), N\rangle = \langle \begin{pmatrix} \dot{x}(0) \\ \dot{y}(0) \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \langle = 0$$

Moreover, for every $0 < s < 1$ we have

$$p + sN = (p_x, p_y, p_z - s) = \left(p_x, p_y, \sqrt{R} - s\right)$$

Where $p_z = \sqrt{R}$ since $p \in D_1$. For every $(x,y,z) \in V_1$ we have $z > \sqrt{R}$ hence $p + sN \notin V_1$ for all $0 < s < 1$.
Consider the map $\varphi : (0, R) \times (0, 2\pi) \to D_1$ defined by $\varphi(t, \theta) = \left(t\cos(\theta), t\sin(\theta), \sqrt{R}\right)$ so we have

$$\frac{\partial \varphi}{\partial t} = \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{pmatrix}, \qquad \frac{\partial \varphi}{\partial \theta} = \begin{pmatrix} -t\sin(\theta) \\ t\cos(\theta) \\ 0 \end{pmatrix}$$

$$\Longrightarrow \langle \frac{\partial \varphi}{\partial t}, \frac{\partial \varphi}{\partial \theta} \rangle = -t\sin(\theta)\cos(\theta) + t\sin(\theta)\cos(\theta) = 0$$

Furthermore

$$\left\| \frac{\partial \varphi}{\partial t} \right\|^2 = \cos^2(\theta) + \sin^2(\theta) = 1$$

$$\left\| \frac{\partial \varphi}{\partial \theta} \right\|^2 = t^2\cos^2(\theta) + t^2\sin^2(\theta) = t^2$$

Hence

$$\begin{aligned}
\int_{D_1} \langle F, N \rangle \,\mathrm{d}\sigma &= \int_{D_1} -(2z^2 + x) \,\mathrm{d}\sigma \\
&= \int_0^{2\pi} \int_0^R -\left(2\left(\sqrt{R}\right)^2 + t\cos(\theta)\right)\sqrt{t^2} \,\mathrm{d}t \,\mathrm{d}\theta \\
&= \int_0^{2\pi} \int_0^R -(2Rt + t^2\cos(\theta)) \,\mathrm{d}t \,\mathrm{d}\theta \\
&= \int_0^{2\pi} \left[-Rt^2 - \frac{t^3}{3}\cos(\theta) \right]_{t=0}^{t=R} \mathrm{d}\theta \\
&= \int_0^{2\pi} \left(-R^3 - \frac{R^3}{3}\cos(\theta) \right) \mathrm{d}\theta \\
&= \left[-R^3\theta - \frac{R^3}{3}\sin(\theta) \right]_{\theta=0}^{\theta=2\pi} \\
&= -2\pi R^3
\end{aligned}$$

For M_1 the required normal N_{up} so

$$\int_{M_1} \langle F, N_{\mathrm{up}} \rangle \,\mathrm{d}\sigma = -\int_{D_1} \langle F, N \rangle \,\mathrm{d}\sigma = 2\pi R^3$$

By identical reasoing for D_2 from V_2 the normal is $N = (0,0,1)$ and we have

$$\int_{D_2} \langle F, N \rangle \,\mathrm{d}\sigma = \int_{D_2} (2z^2 + x) \,\mathrm{d}\sigma = \int_0^{2\pi} \int_0^R \left(2\left(-\sqrt{R}\right)^2 + t\cos(\theta)\right) t \,\mathrm{d}t \,\mathrm{d}\theta = \int_0^{2\pi} \left[Rt^2 + \frac{t^3}{3}\cos(\theta) \right]_{t=0}^{t=R} \mathrm{d}\theta = 2\pi R^3$$

Hence

$$-\int_{M_2} \langle F, N_{\mathrm{up}} \rangle \,\mathrm{d}\sigma + 2\pi R^3 = 0 \Longrightarrow \int_{M_2} \langle F, N_{\mathrm{up}} \rangle \,\mathrm{d}\sigma = 2\pi R^3$$

So the total upward flux is

$$\int_M \langle F, N_{\mathrm{up}} \rangle \,\mathrm{d}\sigma = \int_{M_1} \langle F, N_{\mathrm{up}} \rangle \,\mathrm{d}\sigma + \int_{M_2} \langle F, N_{\mathrm{up}} \rangle \,\mathrm{d}\sigma = 2\pi R^3 + 2\pi R^3 = 4\pi R^3$$

We have $R = \frac{\sqrt{5}-1}{2}$ as the positive root of $R^2 + R + 1 = 0$ so

$$R^3 = R(1-R) = R - R^2 = R - (1-R) = 2R - 1 = \sqrt{5} - 2$$

So the final flux is $4\pi\left(\sqrt{5} - 2\right)$. □